

ELITE IGCSE ACADEMY

Edexcel IGCSE Mathematics A - Unit 1 (4WM1)

Higher Tier

Elite IGCSE Classified Unit 1 Expertise (Q20+)

131 Questions | Unit 1 / 4WM1 | Q20+ Expertise | Question book

classified questions

pathway palette

course scope

CLASSIFIED TOPIC

MODULAR

Set Notation & Venn Diagrams

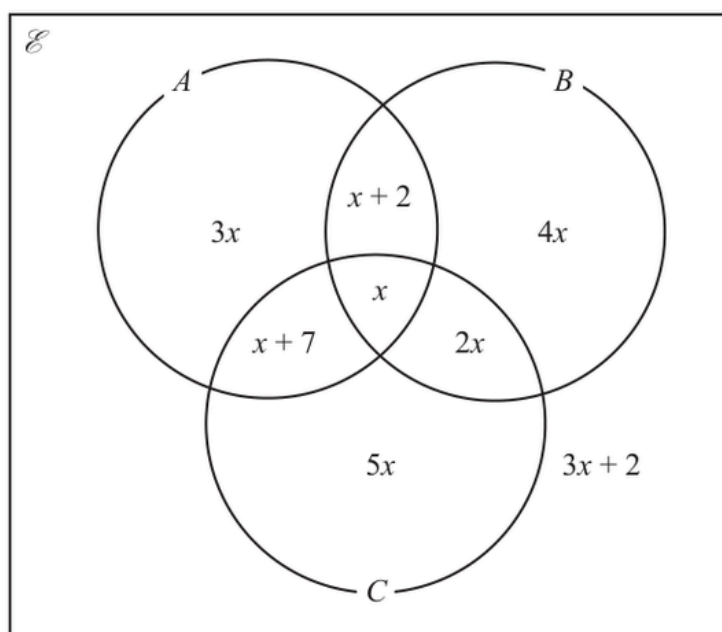
1 questions

Set Notation & Venn Diagrams

Q#: 21

Marks: 4

- 21 The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$n(A' \cap C) = \dots\dots\dots$

(Total for Question 21 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Rounding, Estimation & Bounds

6 questions

Rounding, Estimation & Bounds

Q#: 22

Marks: 4

22 The diagram shows triangle ABC

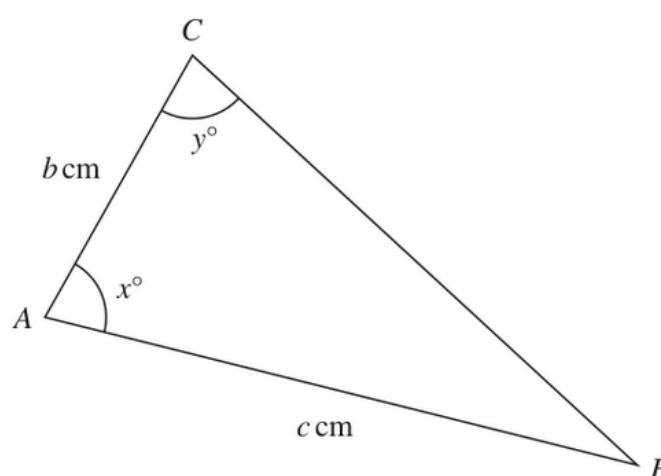


Diagram **NOT**
accurately drawn

$c = 11.5$ correct to one decimal place
 $x = 80$ correct to the nearest whole number
 $y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b

Show your working clearly.

Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

Rounding, Estimation & Bounds

Q#: 25

Marks: 4

- 25** A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

..... g/cm^3

(Total for Question 25 is 4 marks)

Rounding, Estimation & Bounds

Q#: 22

Marks: 4

22 The diagram shows a square inside rectangle $ABCD$

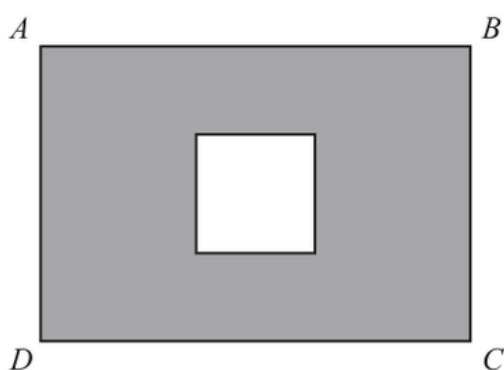


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X\text{cm}^2$

$AB = 11.5\text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2\text{ cm}$ correct to 2 significant figures

side of square $= 4.1\text{ cm}$ correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

Rounding, Estimation & Bounds

Q#: 20

Marks: 3

20 $P = \frac{t - w}{y}$

$t = 9.7$ correct to 1 decimal place

$w = 5.9$ correct to 1 decimal place

$y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P .
Show your working clearly.

(Total for Question 20 is 3 marks)

Rounding, Estimation & Bounds

Q#: 21

Marks: 3

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

(Total for Question 21 is 3 marks)

Rounding, Estimation & Bounds

Q#: 24

Marks: 5

24 $D = \frac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place

$p = 7.24$ correct to 2 decimal places

$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

(Total for Question 24 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Surds

3 questions

Surds

Q#: 20

Marks: 3

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $(\sqrt{7} + 1)\text{ cm}$.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

(Total for Question 20 is 3 marks)

Surds

Surds

Q#: 21

Marks: 4

21 Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

Surds

Q#: 23

Marks: 4

23 The diagram shows a cuboid with a square cross section.

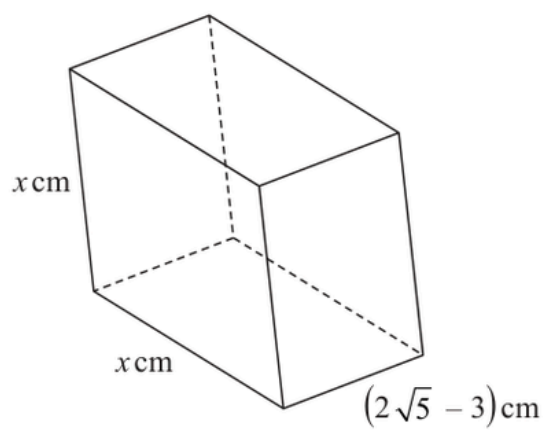


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x =$

(Total for Question 23 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Algebraic Roots & Indices

6 questions

Algebraic Roots & Indices

Q#: 22

Marks: 3

22 $\frac{10 \times (\sqrt[3]{27})}{6 \times 9^{2n+8}} = 3^x$

Express x in terms of n

Show your working clearly and simplify your expression.

$x = \dots\dots\dots$

Algebraic Roots & Indices

Q#: 21

Marks: 7

21 (a) Simplify fully $\frac{10x^2 + 23x + 12}{4x^2 - 9}$

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

(3)

(b) Find an expression for n in terms of y .
Show clear algebraic working and simplify your expression.

(4)

(Total for Question 21 is 7 marks)

Algebraic Roots & Indices

Q#: 24

Marks: 3

24

$$\frac{18 \times (\sqrt{27})}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

(Total for Question 24 is 3 marks)

Algebraic Roots & Indices

Q#: 23

Marks: 3

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

(Total for Question 23 is 3 marks)

Algebraic Roots & Indices

Q#: 24

Marks: 5

24 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n State any restrictions on n **(Total for Question 24 is 5 marks)**

Algebraic Roots & Indices

Q#: 21

Marks: 5

21 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

(Total for Question 21 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Completing the Square

13 questions

Completing the Square

Completing the Square

Q#: 25

Marks: 5

- 25 (a) Express $8 - 12x - 4x^2$ in the form $a - b(x + c)^2$ where a , b and c are constants.

(3)

- (b) Hence solve $8 - 12x - 4x^2 = 0$
Show your working clearly.
Give your solutions in surd form.

(2)

(Total for Question 25 is 5 marks)

Completing the Square

Completing the Square

Q#: 22

Marks: 2

22 $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where p , q and r are integers.

Find the value of q

$q = \dots\dots\dots$

(Total for Question 22 is 2 marks)

Completing the Square

Q#: 24

Marks: 3

24 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 24 is 3 marks)

Completing the Square

Q#: 20

Marks: 4

20 Find the values of a , b and c so that

$$7 + 12x - 2x^2$$

is written as $a - b(x - c)^2$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 20 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Algebraic Fractions

16 questions

Algebraic Fractions

Q#: 20

Marks: 3

20 Given that $k = m + n$ and $m = \frac{3}{4n}$

express $\frac{7k}{5-m}$ in the form $\frac{a+bn^2}{cn-d}$ where a, b and c are integers and d is prime.

(Total for Question 20 is 3 marks)

Algebraic Fractions

Q#: 24

Marks: 4

24 Write $(2x - 1) + \frac{2x^2 - 5x - 12}{x^2 - 16} \times \frac{x^2 + 4x}{6x^2 + 11x + 3}$ as a single fraction in its simplest form.

Show clear algebraic working.

(Total for Question 24 is 4 marks)

Algebraic Fractions

Q#: 25

Marks: 3

25 $a = \frac{2x + 5}{1 - x} \quad x = \frac{5 - 2y}{3y}$

Write a in the form $\frac{m + ny}{p(y - 1)}$ where m , n and p are integers.

Show your working clearly.

$a =$

(Total for Question 25 is 3 marks)

Algebraic Fractions

Q#: 28

Marks: 4

28 Solve $\frac{6x^3 + 5x^2 + x}{4x^2 - 1} \div \frac{15x^2 - x - 2}{8x - 4} = 2x$

Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 28 is 4 marks)

Algebraic Fractions

Q#: 21

Marks: 3

- 21** Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.
Show clear algebraic working.

CLASSIFIED TOPIC

MODULAR

Completing the Square

13 questions

Completing the Square

Completing the Square

Q#: 23

Marks: 3

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

(Total for Question 23 is 3 marks)

Completing the Square

Q#: 20

Marks: 4

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point *A* is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of *A*

(.....,)
(1)

(Total for Question 20 is 4 marks)

Completing the Square

Q#: 24

Marks: 4

24 Express each of a , b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

Completing the Square

Q#: 23

Marks: 5

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

.....
(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(..... ,)
(2)

(Total for Question 23 is 5 marks)

Completing the Square

Q#: 25

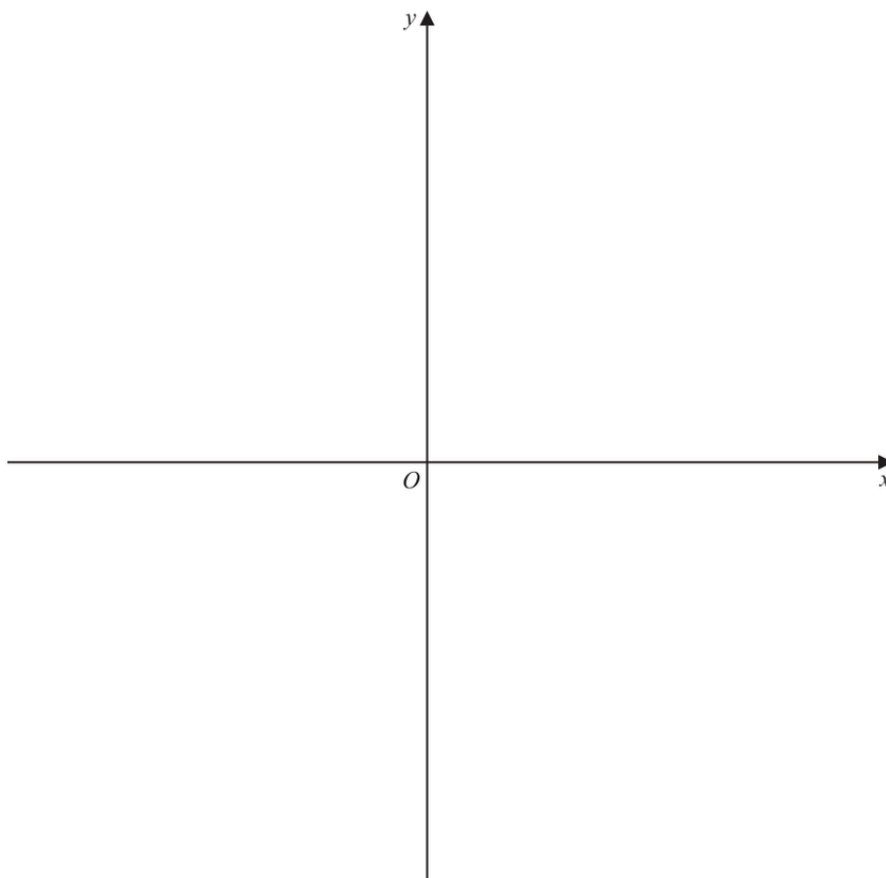
Marks: 6

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

(3)

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(3)

(Total for Question 25 is 6 marks)

Completing the Square

Q#: 22

Marks: 4

- 22** The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants,
describe fully a series of transformations that map the curve **S** onto the curve **T**.

(Total for Question 22 is 4 marks)

Completing the Square

Q#: 25

Marks: 4

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants.

$$f(x) = \dots\dots\dots$$

(Total for Question 25 is 4 marks)

Completing the Square

Completing the Square

Q#: 20

Marks: 5

- 20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a , b and c are numbers to be found.

.....
(3)

The curve C has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on C

- (b) Find the coordinates of P

(..... ,)
(2)

(Total for Question 20 is 5 marks)

Completing the Square

Q#: 21

Marks: 3

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 21 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Algebraic Fractions

16 questions

Algebraic Fractions

Q#: 24

Marks: 4

24 Express

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

(Total for Question 24 is 4 marks)

Algebraic Fractions

Q#: 21

Marks: 4

21 Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

(Total for Question 21 is 4 marks)

Algebraic Fractions

Q#: 23

Marks: 3

23 Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

(Total for Question 23 is 3 marks)

Algebraic Fractions

Q#: 20

Marks: 3

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x + 2}$ in the form $\frac{a - by^2}{c + dy}$ where a, b, c and d are integers.

(Total for Question 20 is 3 marks)

Algebraic Fractions

Q#: 26

Marks: 4

26 Write $4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

Algebraic Fractions

Q#: 23

Marks: 4

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p = \dots\dots\dots$

(Total for Question 23 is 4 marks)

Algebraic Fractions

Q#: 21

Marks: 4

21 Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a .

Give your expression as a single fraction in its simplest form.

(Total for Question 21 is 4 marks)

Algebraic Fractions

Q#: 23

Marks: 4

23 Simplify $(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x} \right) - 2x$

Give your answer in the form $\frac{ax^2}{bx + c}$ where a , b and c are integers.

(Total for Question 23 is 4 marks)

Algebraic Fractions

Q#: 21

Marks: 5

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

Algebraic Fractions

Q#: 24

Marks: 5

24 Solve $\frac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 24 is 5 marks)

Algebraic Fractions

Q#: 23

Marks: 3

23 $a = \frac{2x + 5}{1 - x} \quad x = \frac{5 - 2y}{3y}$

Write a in the form $\frac{m + ny}{p(y - 1)}$ where m , n and p are integers.

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 23 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Coordinate Geometry

24 questions

Coordinate Geometry

Q#: 24

Marks: 5

24 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates $(4, 7)$

R is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points Q and S

Give your answer in the form $ay = bx + c$ where a , b and c are integers.

(Total for Question 24 is 5 marks)

Coordinate Geometry

Q#: 23

Marks: 5

23 A circle C has centre $(3q, 4q)$ where q is an integer.

The straight line with equation $y = -\frac{5}{4}x + \frac{21}{4}$ is the tangent to C at the point $(p, p + 3)$ where p is an integer.

Work out the radius of C

Give your answer in the form $\sqrt{a}$ where a is an integer.

Show your working clearly.

(Total for Question 23 is 5 marks)

Coordinate Geometry

Q#: 27

Marks: 5

27 $ABCD$ is a rhombus with $AB = BC = CD = DA$

A has coordinates $(16, -9)$

C has coordinates $(24, 15)$

Find an equation of the line BD

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

Show your working clearly.

(Total for Question 27 is 5 marks)

Coordinate Geometry

Q#: 23

Marks: 6

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.

The point D has coordinates $(8, 7)$

The straight line L passes through the points B and D and has a gradient of $\frac{3}{5}$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.

(Total for Question 23 is 6 marks)

Coordinate Geometry

Q#: 22

Marks: 6

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 22 is 6 marks)

Coordinate Geometry

Q#: 22

Marks: 5

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B .

Find the coordinates of A and B .
Show clear algebraic working.

(..... ,)

(..... ,)

(Total for Question 22 is 5 marks)

Coordinate Geometry

Q#: 24

Marks: 5

24 L_1 and L_2 are two straight lines.

The origin of the coordinate axes is O .

L_1 has equation $5x + 10y = 8$

L_2 is perpendicular to L_1 and passes through the point with coordinates $(8, 6)$

L_2 crosses the x -axis at the point A .

L_2 intersects the straight line with equation $x = -3$ at the point B .

Find the area of triangle AOB .

Show your working clearly.

(Total for Question 24 is 5 marks)

Coordinate Geometry

Q#: 22

Marks: 5

22 ABC is an isosceles triangle with $AB = AC$.

B is the point with coordinates $(-1, 5)$

C is the point with coordinates $(2, 10)$

M is the midpoint of BC .

Find an equation of the line through the points A and M .

Give your answer in the form $py + qx = r$ where p, q and r are integers.

(Total for Question 22 is 5 marks)

Coordinate Geometry

Q#: 22

Marks: 5

22 ABC is a triangle in which angle $ABC = 90^\circ$

p and q are integers such that

the coordinates of A are $(p, 10)$

the coordinates of B are $(-1, -5)$

the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(Total for Question 22 is 5 marks)

Coordinate Geometry

Q#: 22

Marks: 7

22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,
find the coordinates of the points B and D .

(..... ,)

(..... ,)

(Total for Question 22 is 7 marks)

Coordinate Geometry

Q#: 20

Marks: 4

20 The centre O of a circle has coordinates $(4, 7)$

The point A , on the circle, has coordinates $(6, 11)$ and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

(Total for Question 20 is 4 marks)

Coordinate Geometry

Q#: 24

Marks: 6

24 $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(.....,)

(Total for Question 24 is 6 marks)

Coordinate Geometry

Q#: 21

Marks: 5

21 Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(..... ,)

(..... ,)

(Total for Question 21 is 5 marks)

Coordinate Geometry

Q#: 25

Marks: 6

- 25** The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line AB where A is the point with coordinates $(j, 7)$ and B is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line AB
Show clear algebraic working.

(..... ,)

(Total for Question 25 is 6 marks)

Coordinate Geometry

Q#: 22

Marks: 5

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(..... ,) (..... ,)

(Total for Question 22 is 5 marks)

Coordinate Geometry

Q#: 21

Marks: 4

21 $ABCD$ is a square.

The point A has coordinates $(-5, 2)$

The point B has coordinates $(3, 5)$

Find an equation of the line that passes through B and C

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(Total for Question 21 is 4 marks)

Coordinate Geometry

Q#: 25

Marks: 5

25 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates $(4, 7)$

R is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points Q and S
Give your answer in the form $ay = bx + c$ where a , b and c are integers.

(Total for Question 25 is 5 marks)

Coordinate Geometry

Q#: 24

Marks: 6

24 PQ is a straight line drawn on a square grid, with a scale of 1 cm for 1 unit on each axis.

P has coordinates $(-5, a)$ and Q has coordinates $(7, 3a)$ where $a > 0$

The length of PQ is $4\sqrt{10}$ cm

Find an equation of the perpendicular bisector of PQ

Give your answer in the form $y = mx + c$ where m and c are integers.

(Total for Question 24 is 6 marks)

Coordinate Geometry

Q#: 22

Marks: 4

22 $ABCD$ is a rhombus.

The diagonals, AC and BD , intersect at the point M .

The coordinates of M are $(6, -11)$

The points A and C both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through B and D intersects the y -axis.

(..... ,)

(Total for Question 22 is 4 marks)

Coordinate Geometry

Q#: 20

Marks: 7

- 20** The straight line **L** passes through point $A(-6, 2)$ and point $B(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B .
The line **M** intersects the line $x = -1$ at point C .

Calculate the area of triangle ABC .

(Total for Question 20 is 7 marks)

Coordinate Geometry

Q#: 22

Marks: 6

22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

Coordinate Geometry

Q#: 21

Marks: 5

- 21** The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(..... ,)

(..... ,)

(Total for Question 21 is 5 marks)

Coordinate Geometry

Q#: 20

Marks: 3

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

Coordinate Geometry

Q#: 28

Marks: 6

- 28** The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are $(0.2, 2.6)$

Work out the coordinates of B
Show your working clearly.

(..... ,)

(Total for Question 28 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Graphs of Functions

7 questions

Graphs of Functions

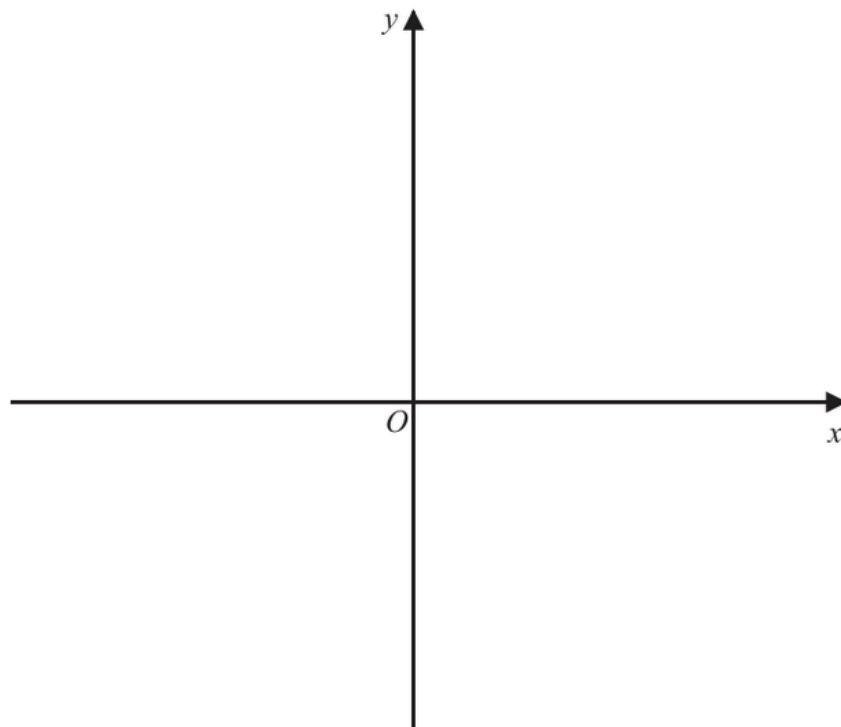
Q#: 20

Marks: 4

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



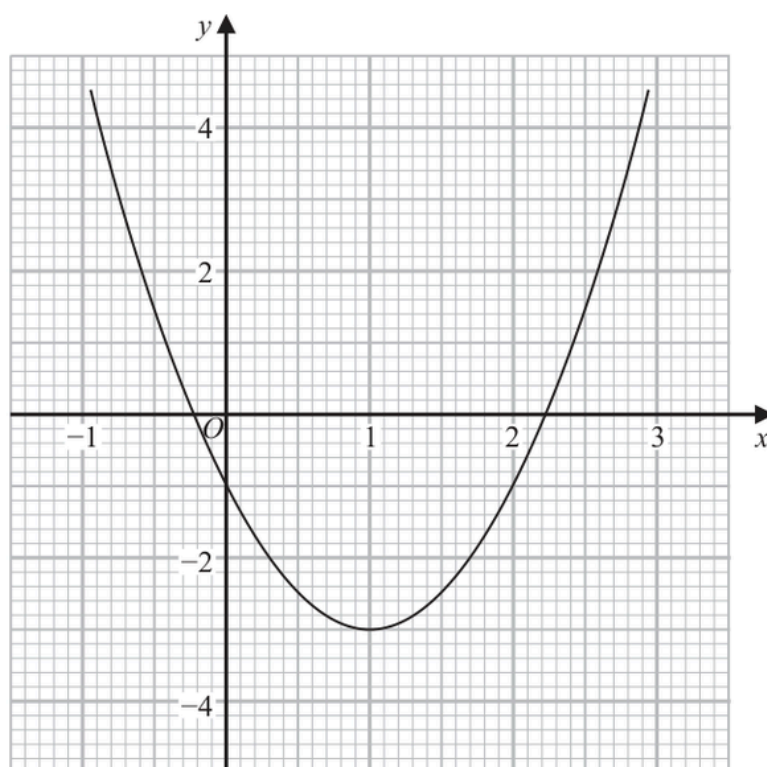
(Total for Question 20 is 4 marks)

Graphs of Functions

Q#: 21

Marks: 5

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

.....
(2)

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....
(3)

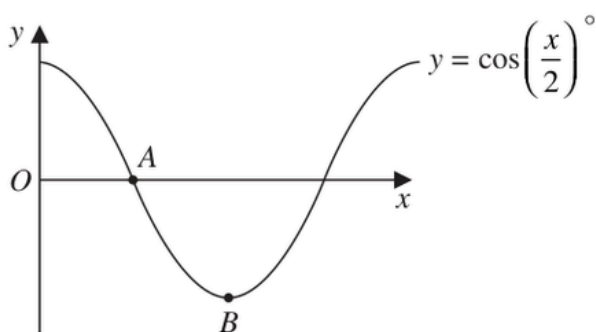
(Total for Question 21 is 5 marks)

Graphs of Functions

Q#: 23

Marks: 2

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

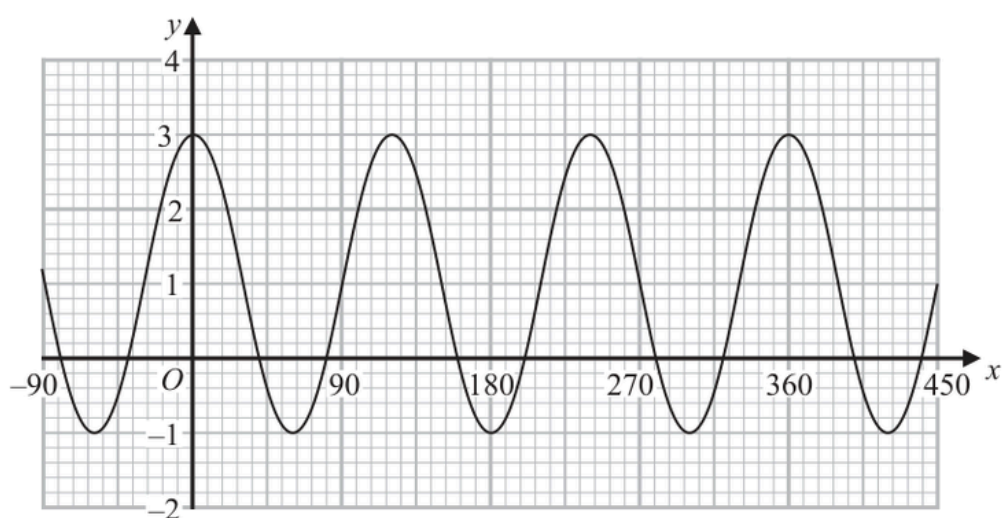
(Total for Question 23 is 2 marks)

Graphs of Functions

Q#: 26

Marks: 3

26 Here is a sketch of the curve with equation $y = a \cos bx^\circ + c$ where $-90 \leq x \leq 450$



Find the value of a , the value of b and the value of c

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$

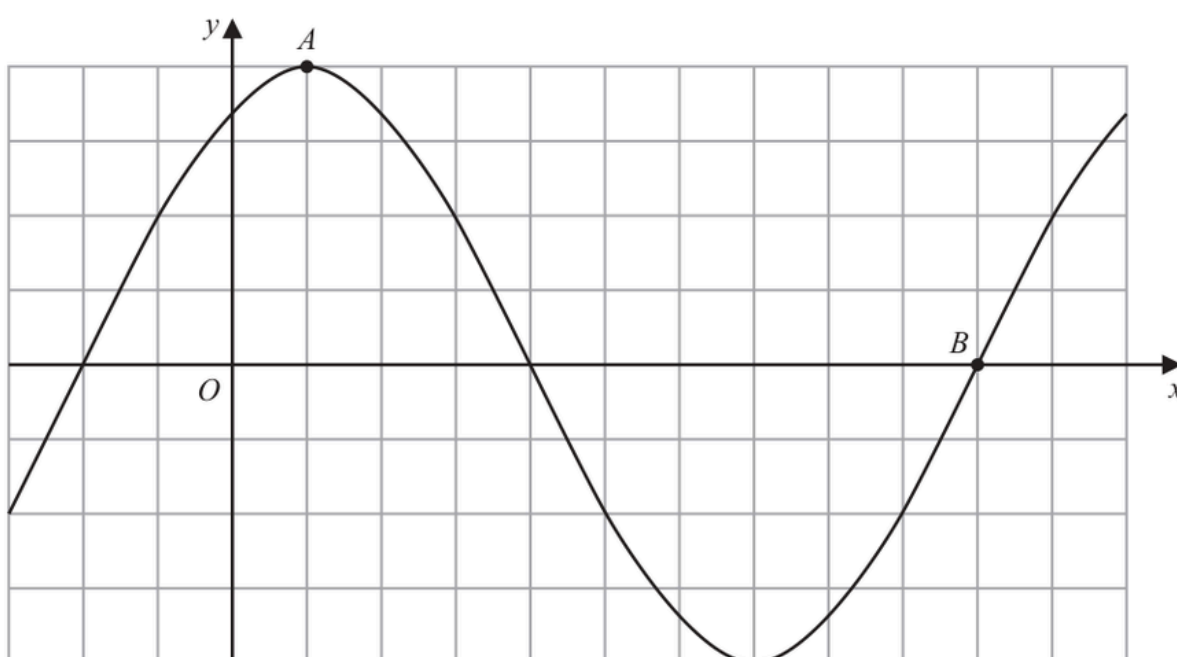
(Total for Question 26 is 3 marks)

Graphs of Functions

Q#: 25

Marks: 2

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point A

(..... ,)
(1)

(ii) Find the coordinates of the point B

(..... ,)
(1)

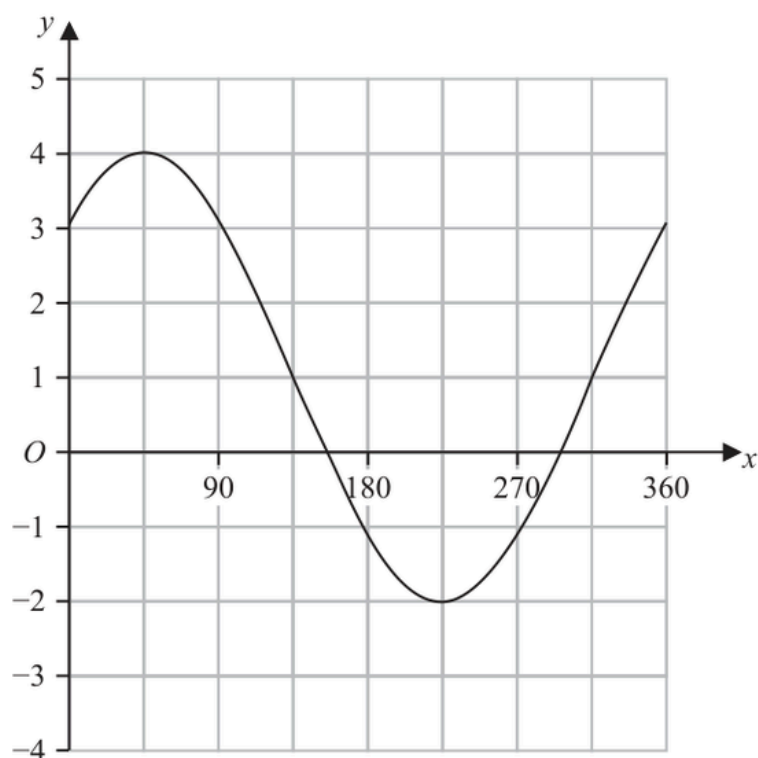
(Total for Question 25 is 2 marks)

Graphs of Functions

Q#: 25

Marks: 3

25 The graph of $y = a \sin(x + b)^\circ + c$ for $0 \leq x \leq 360$ is drawn on the grid below.



Find a suitable value for a , for b and for c

$a =$

$b =$

$c =$

(Total for Question 25 is 3 marks)

Graphs of Functions

Q#: 23

Marks: 5

- 23** The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.
- C** and **L** intersect at the points A and B
- The coordinates of point A are (p, k)
The coordinates of point B are (q, k)
- Given that $p - q = 14$
- find the value of k
Show clear algebraic working.

 $k = \dots\dots\dots$ **(Total for Question 23 is 5 marks)**

CLASSIFIED TOPIC

MODULAR

Standard & Compound Units

2 questions

Standard & Compound Units

Q#: 21

Marks: 6

- 21 The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

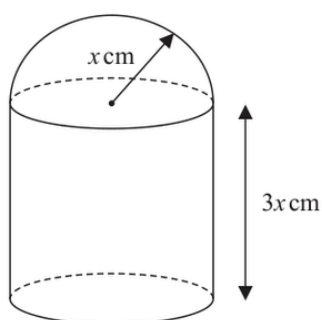


Diagram **NOT**
accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both $x \text{ cm}$.
The height of the cylinder is $3x \text{ cm}$.

The total surface area of the solid is $81\pi \text{ cm}^2$
The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

Metal	Density (g/cm^3)
Aluminium	2.7
Nickel	8.9
Gold	19.3
Silver	10.5

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.
Show your working clearly.

(Total for Question 21 is 6 marks)

Standard & Compound Units

Q#: 26

Marks: 6

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

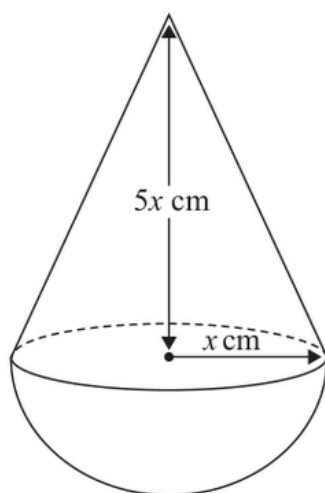


Diagram **NOT**
accurately drawn

The cone is made from copper.
The density of copper is 9 g/cm^3

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams
The total volume of the solid is $504\pi \text{ cm}^3$

Work out the density of the hemisphere.
Show your working clearly.

..... g/cm^3

(Total for Question 26 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Area & Perimeter

3 questions

Area & Perimeter

Q#: 21

Marks: 4

- 21 The diagram shows an equilateral triangle and a circle.

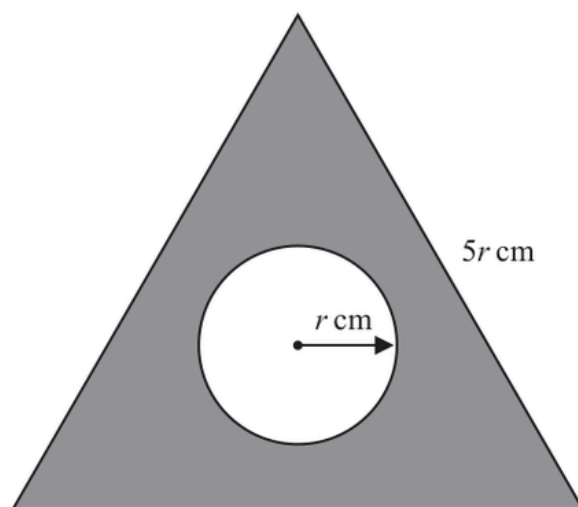


Diagram **NOT**
accurately drawn

The equilateral triangle has sides of length $5r \text{ cm}$
The circle has radius $r \text{ cm}$

The area of the shaded region is $610\pi \text{ cm}^2$
Work out the value of r

Give your answer correct to 3 significant figures.

$r = \dots\dots\dots$

(Total for Question 21 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Circles, Arcs & Sectors

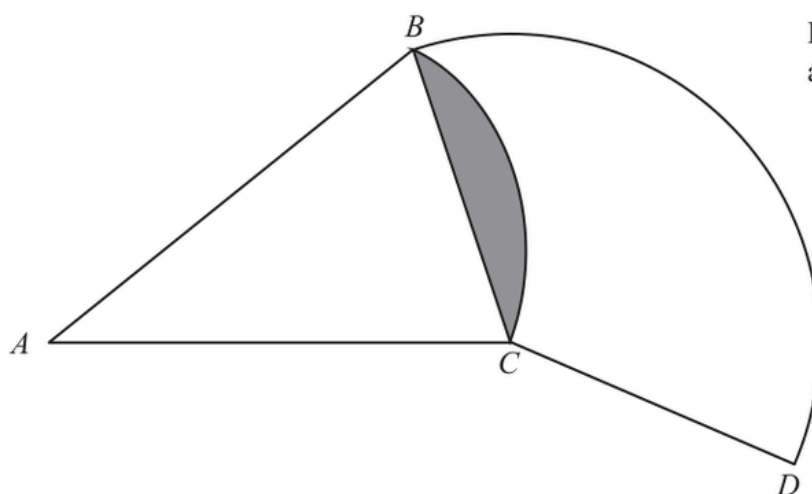
11 questions

Circles, Arcs & Sectors

Q#: 22

Marks: 6

22

Diagram **NOT**
accurately drawn BAC is a sector of a circle, centre A BCD is a sector of a circle, centre C Angle $BAC = 40^\circ$ Angle $BCD = 130^\circ$ Area of shaded segment = 28 cm^2 Find the length of the arc BD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Area & Perimeter

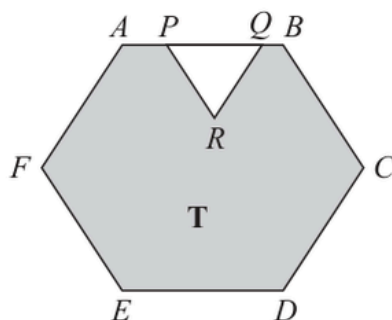
3 questions

Area & Perimeter

Q#: 20

Marks: 4

20

Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

..... cm

(Total for Question 20 is 4 marks)

Area & Perimeter

Q#: 22

Marks: 6

22 The diagram shows a regular octagon $ABCDEFGH$.

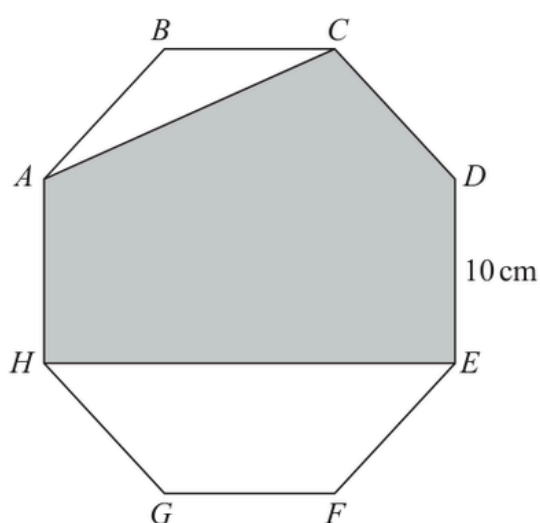


Diagram **NOT**
accurately drawn

Each side of the octagon has length 10 cm .

Find the area of the shaded region $ACDEH$.
Give your answer correct to the nearest cm^2

..... cm^2

(Total for Question 22 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Circles, Arcs & Sectors

11 questions

Circles, Arcs & Sectors

Q#: 22

Marks: 6

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

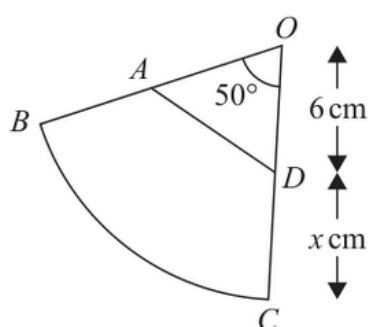


Diagram **NOT**
accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$

(Total for Question 22 is 6 marks)

Circles, Arcs & Sectors

Q#: 20

Marks: 5

20 A , B and C are points on a circle with centre O .

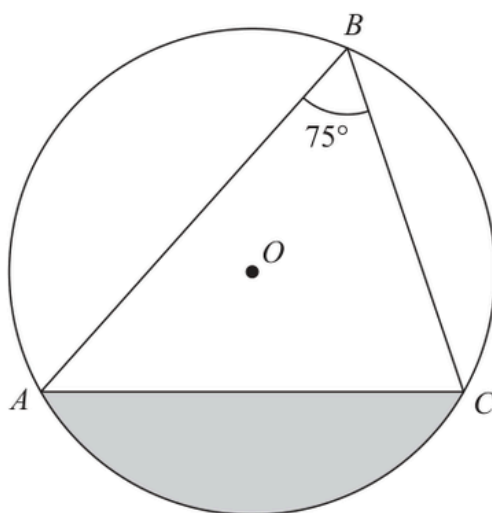


Diagram **NOT**
accurately drawn

Angle $ABC = 75^\circ$

The area of the shaded segment is 200 cm^2

Calculate the radius of the circle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Circles, Arcs & Sectors

Q#: 20

Marks: 4

20 The diagram shows four identical circles drawn inside a square.

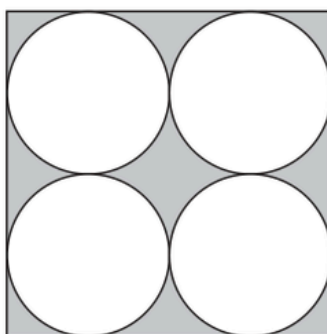


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 4 marks)

Circles, Arcs & Sectors

Q#: 21

Marks: 5

- 21 The diagram shows a square $ABCD$ and a circle.

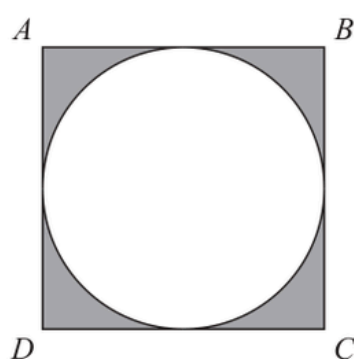


Diagram **NOT**
accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of AC

Give your answer correct to 3 significant figures.

..... cm

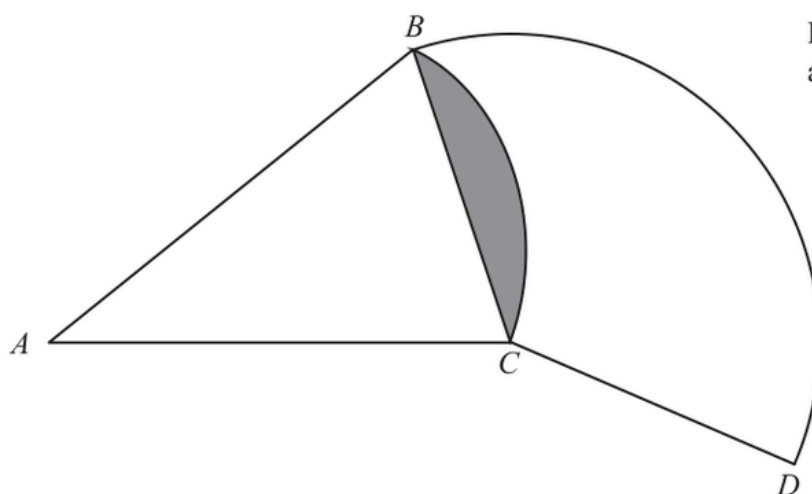
(Total for Question 21 is 5 marks)

Circles, Arcs & Sectors

Q#: 25

Marks: 6

25

Diagram **NOT**
accurately drawn

BAC is a sector of a circle, centre A

BCD is a sector of a circle, centre C

Angle $BAC = 40^\circ$

Angle $BCD = 130^\circ$

Area of shaded segment = 28 cm^2

Find the length of the arc BD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 25 is 6 marks)

Circles, Arcs & Sectors

Q#: 25

Marks: 6

- 25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

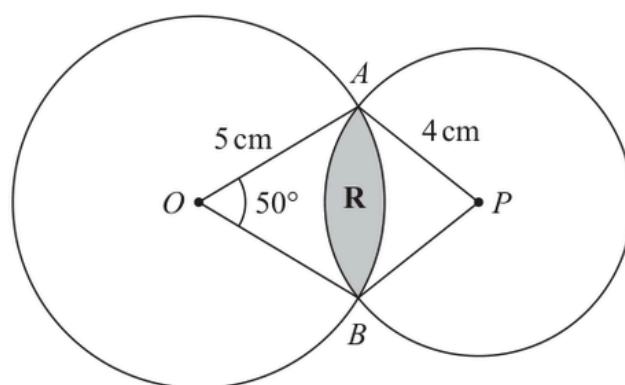


Diagram **NOT**
accurately drawn

One of the circles has centre O and radius 5 cm .

The other circle has centre P and radius 4 cm .

Angle $AOB = 50^\circ$

Calculate the area of region **R**.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 25 is 6 marks)

Circles, Arcs & Sectors

Q#: 22

Marks: 6

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

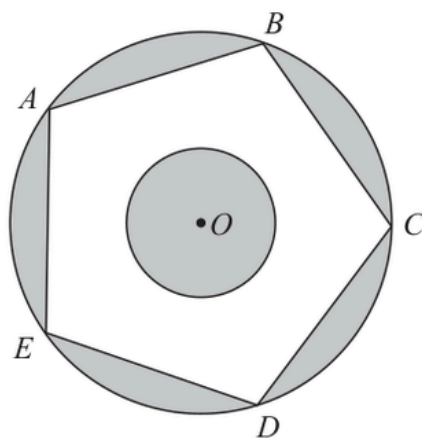


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

Circles, Arcs & Sectors

Q#: 20

Marks: 4

20 The diagram shows a sector $OABC$ of a circle centre O

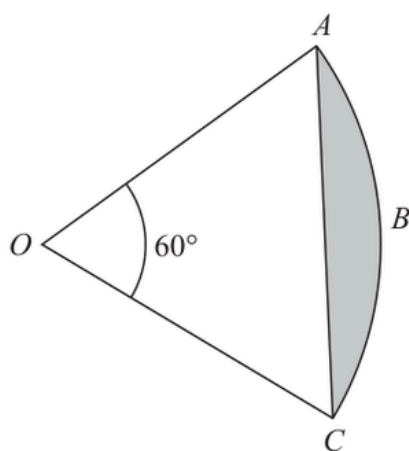


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

..... cm

(Total for Question 20 is 4 marks)

Circles, Arcs & Sectors

Q#: 20

Marks: 4

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm. A circle is drawn inside the triangle.

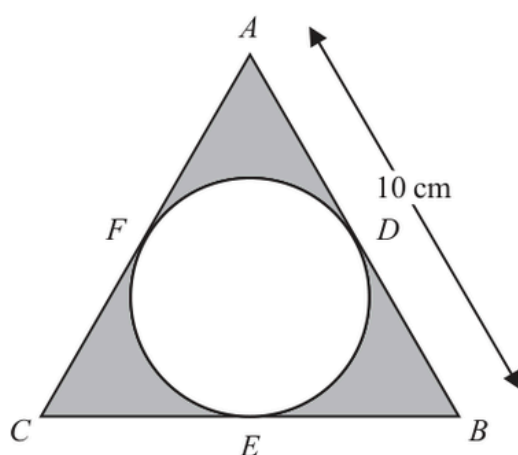


Diagram **NOT**
accurately drawn

D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 20 is 4 marks)

Circles, Arcs & Sectors

Q#: 25

Marks: 5

25 The diagram shows an equilateral triangle ABC and a circle with centre O

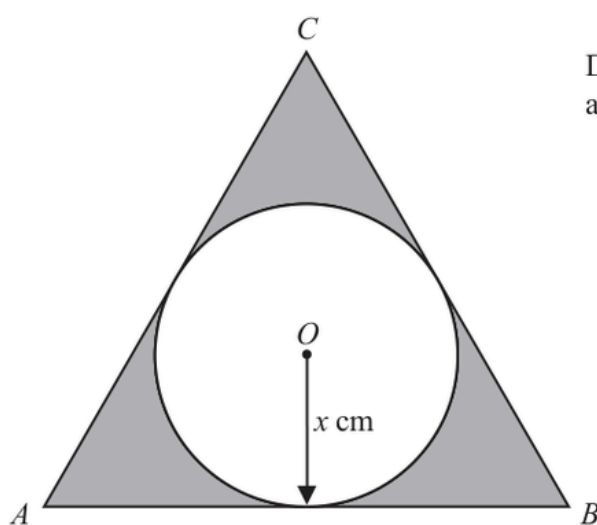


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

$n = \dots\dots\dots$

(Total for Question 25 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Sine, Cosine Rule & Area of Triangles

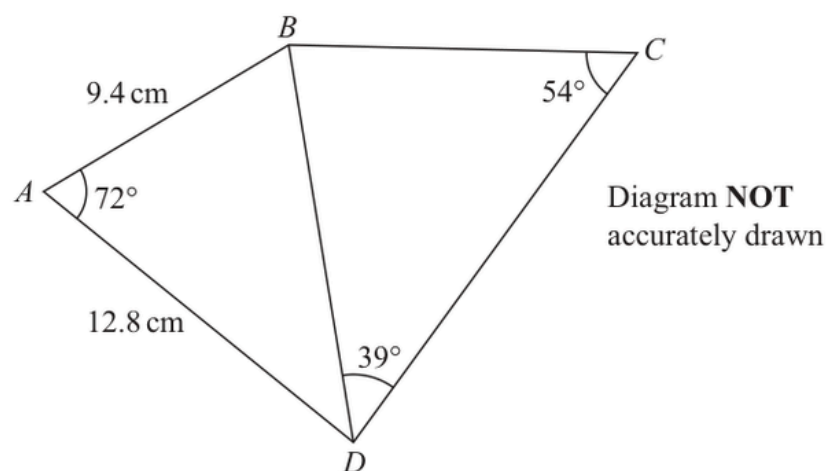
6 questions

Sine, Cosine Rule & Area of Triangles

Q#: 21

Marks: 5

21



Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 21 is 5 marks)

CLASSIFIED TOPIC

MODULAR

3D Pythagoras & Trigonometry

12 questions

3D Pythagoras & Trigonometry

Q#: 23

Marks: 6

23 The diagram shows cuboid $ABCDEFGH$

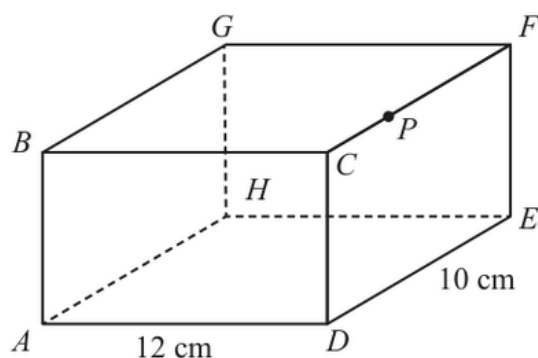


Diagram **NOT**
accurately drawn

The point P lies on CF

$$AD = 12 \text{ cm} \quad DE = 10 \text{ cm} \quad CP = 3 \text{ cm}$$

The angle of elevation of P from A is 24°

Calculate the size of angle APG

Give your answer correct to the nearest degree.

Show your working clearly.

(Total for Question 23 is 6 marks)

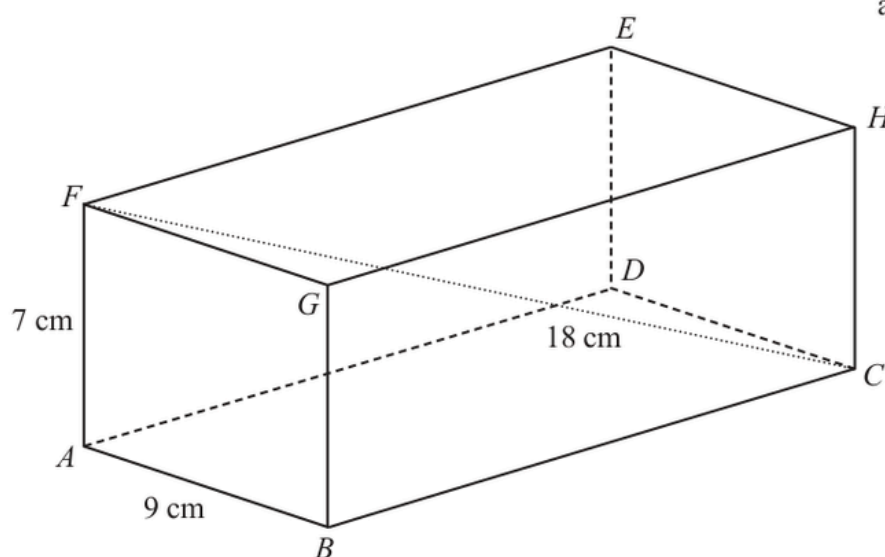
3D Pythagoras & Trigonometry

Q#: 20

Marks: 3

20 The diagram shows cuboid $ABCDEFGH$

Diagram **NOT**
accurately drawn



$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

Calculate the length of BC

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 3 marks)

3D Pythagoras & Trigonometry

Q#: 26

Marks: 5

26 The diagram shows prism $ABCDEFGH$

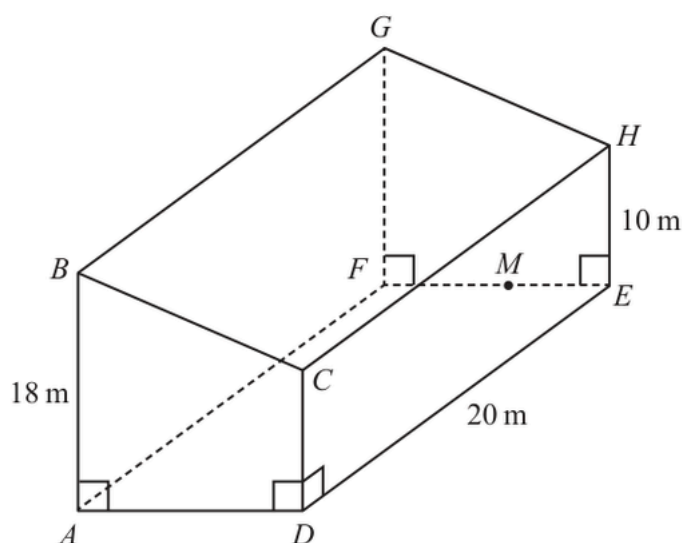


Diagram **NOT**
accurately drawn

$$AB = FG = 18\text{ m} \quad DC = EH = 10\text{ m} \quad AD = FE$$

$$DE = CH = AF = BG = 20\text{ m}$$

$$\text{angle } BAD = \text{angle } ADC = \text{angle } CDE = 90^\circ$$

M is the point on FE such that $FM = \frac{3}{5}FE$

Given that angle $GMF = 60^\circ$

calculate the angle of elevation of G from D

Give your answer correct to one decimal place.

Show your working clearly.

(Total for Question 26 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Sine, Cosine Rule & Area of Triangles

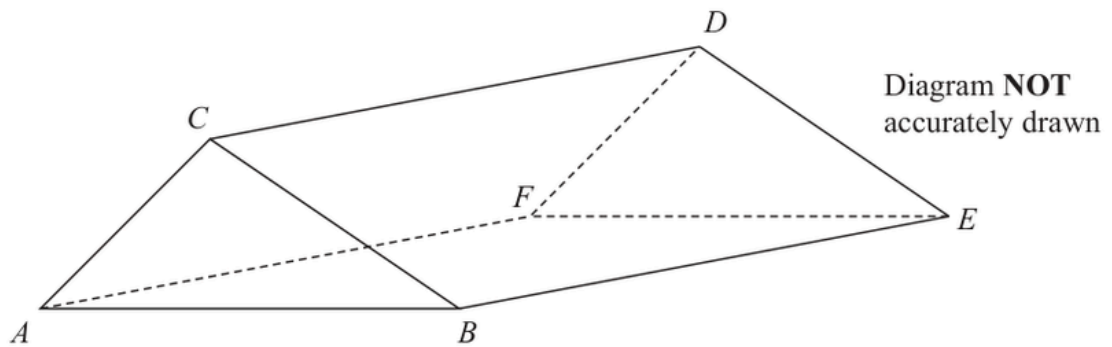
6 questions

Sine, Cosine Rule & Area of Triangles

Q#: 21

Marks: 6

21 The diagram shows the prism $ABCDEF$ with cross section triangle ABC .



Angle $BEC = 40^\circ$ and angle ACB is obtuse.
 $AC = 6\text{ cm}$ and $CE = 13\text{ cm}$

The area of triangle ABC is 22 cm^2

Calculate the length of AB .
 Give your answer correct to one decimal place.

..... cm

(Total for Question 21 is 6 marks)

Sine, Cosine Rule & Area of Triangles

Q#: 23

Marks: 6

23 The diagram shows triangle PQR

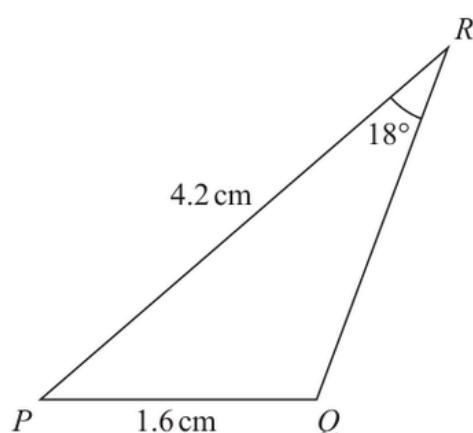


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 23 is 6 marks)

Sine, Cosine Rule & Area of Triangles

Q#: 22

Marks: 5

22 The diagram shows a triangle ABC and a flagpole BF

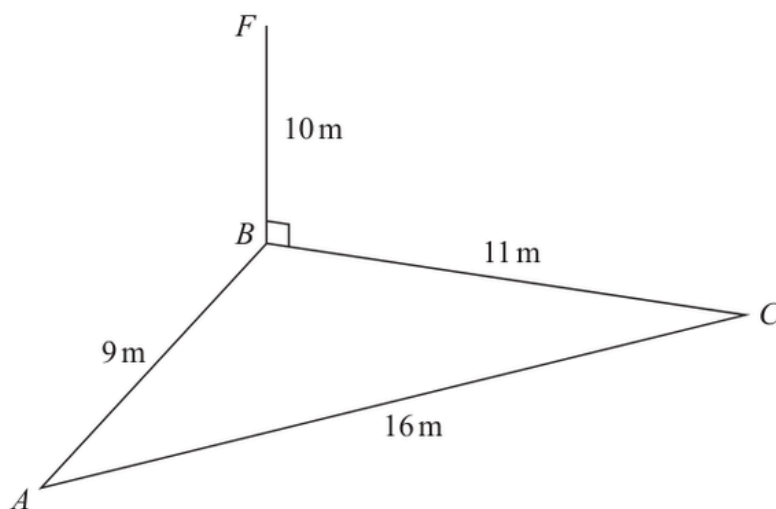


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9\text{ m} \quad BC = 11\text{ m} \quad AC = 16\text{ m} \quad BF = 10\text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D

Give your answer correct to one decimal place.

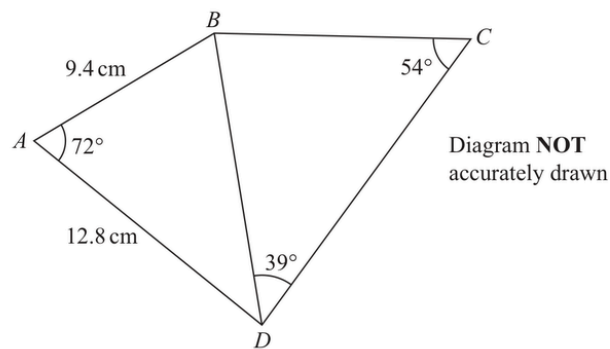
(Total for Question 22 is 5 marks)

Sine, Cosine Rule & Area of Triangles

Q#: 20

Marks: 5

20



Work out the length of BC
 Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Sine, Cosine Rule & Area of Triangles

Q#: 25

Marks: 6

25 Here is a triangle ABC

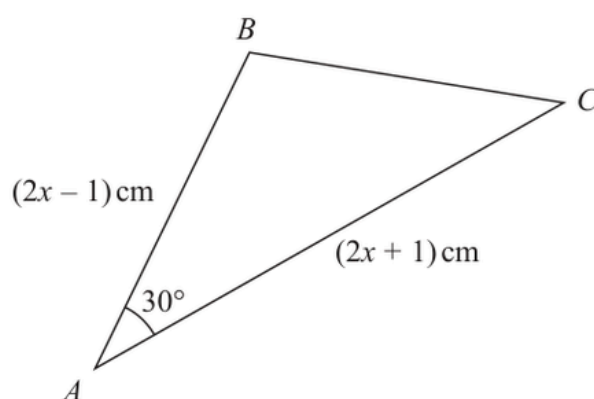


Diagram **NOT**
accurately drawn

The area of the triangle is $(x^2 + x - 3.75) \text{ cm}^2$

Find the size of the largest angle in triangle ABC
Give your answer correct to the nearest degree.

(Total for Question 25 is 6 marks)

CLASSIFIED TOPIC

MODULAR

3D Pythagoras & Trigonometry

12 questions

3D Pythagoras & Trigonometry

Q#: 22

Marks: 4

- 22 ABC is an isosceles triangle in a horizontal plane.
The point T is vertically above B .

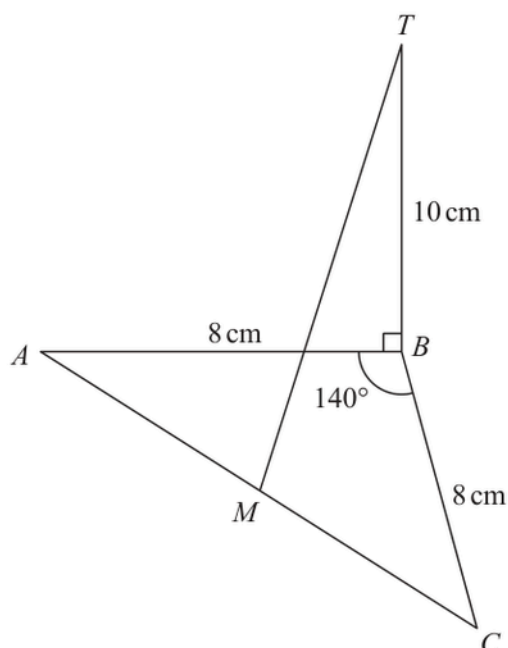


Diagram **NOT**
accurately drawn

Angle $ABC = 140^\circ$
 $AB = BC = 8 \text{ cm}$
 $TB = 10 \text{ cm}$
 M is the midpoint of AC .

Calculate the size of the angle between MT and the horizontal plane ABC .
 Give your answer correct to one decimal place.

(Total for Question 22 is 4 marks)

3D Pythagoras & Trigonometry

Q#: 21

Marks: 5

21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

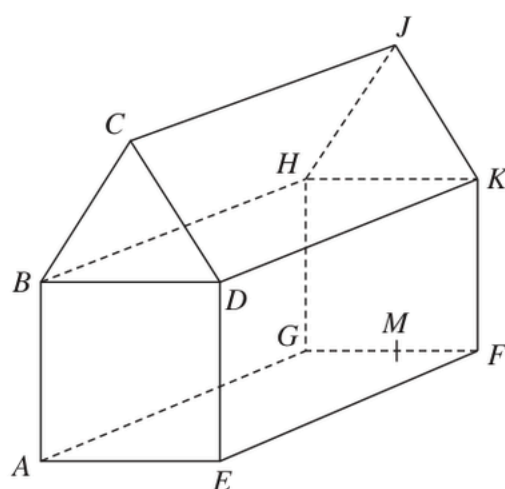


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y^\circ$

Given that $\tan y^\circ = T$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

(Total for Question 21 is 5 marks)

3D Pythagoras & Trigonometry

Q#: 22

Marks: 5

22 Here is a cuboid $ABCDEFGH$

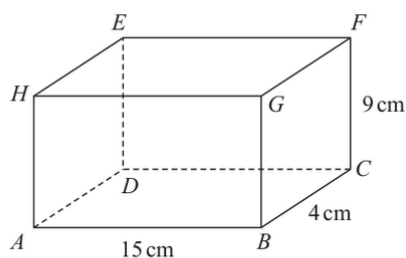


Diagram **NOT**
accurately drawn

$$AB = 15 \text{ cm} \quad BC = 4 \text{ cm} \quad CF = 9 \text{ cm}$$

- (a) Work out the length of BE
Give your answer correct to 3 significant figures.

..... cm
(2)

Here is a cuboid $PQRSTUVW$

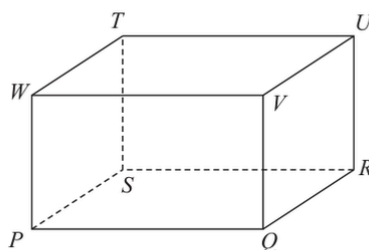


Diagram **NOT**
accurately drawn

$$PR = 42 \text{ cm}$$

The size of the angle between PU and the plane $PQRS$ is 30°

M is the midpoint of PR

- (b) Work out the size of angle UMR
Give your answer correct to 3 significant figures.

.....
(3)

(Total for Question 22 is 5 marks)

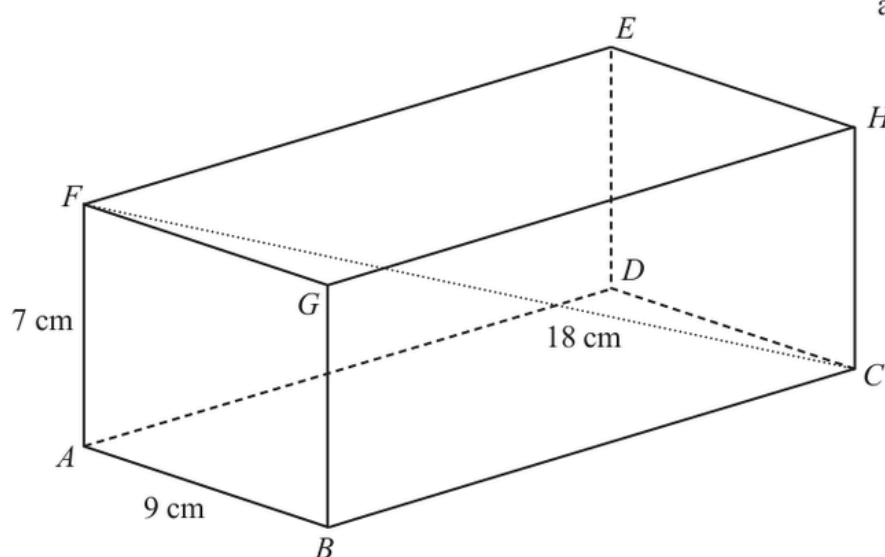
3D Pythagoras & Trigonometry

Q#: 20

Marks: 3

20 The diagram shows cuboid $ABCDEFGH$

Diagram **NOT**
accurately drawn



$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

Calculate the length of BC

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 3 marks)

3D Pythagoras & Trigonometry

Q#: 24

Marks: 4

24 The diagram shows a square-based pyramid $ABCDE$

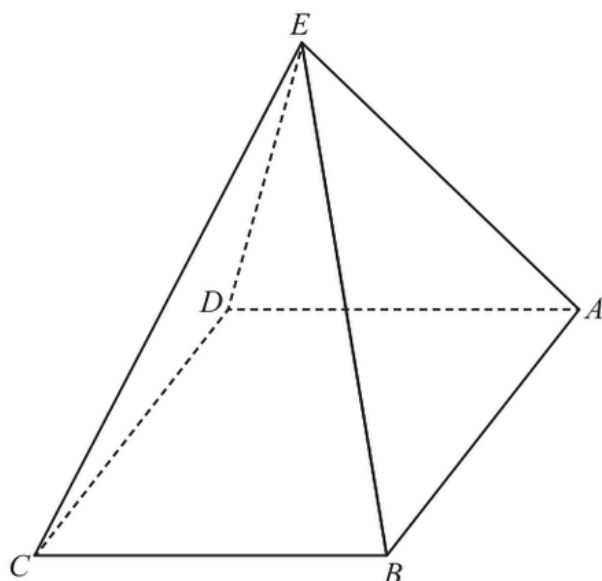


Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

M is the centre of the horizontal square base $ABCD$

Q is the midpoint of AB

$$\text{Angle } EQM = 80^\circ$$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

3D Pythagoras & Trigonometry

Q#: 23

Marks: 3

- 23 The diagram shows a triangular prism, $ABCDEF$, with a horizontal rectangular base $ABCD$

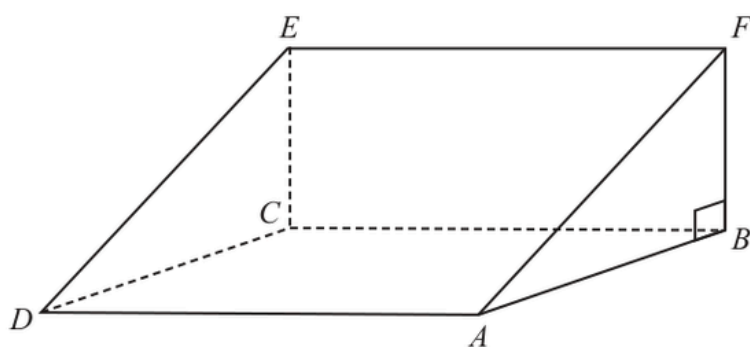


Diagram **NOT**
accurately drawn

M is the midpoint of the line AC

$AC = 40 \text{ cm}$ angle $CAE = 35^\circ$ angle $ABF = 90^\circ$

Work out the size of angle CME

Give your answer correct to 3 significant figures.

(Total for Question 23 is 3 marks)

3D Pythagoras & Trigonometry

Q#: 22

Marks: 4

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

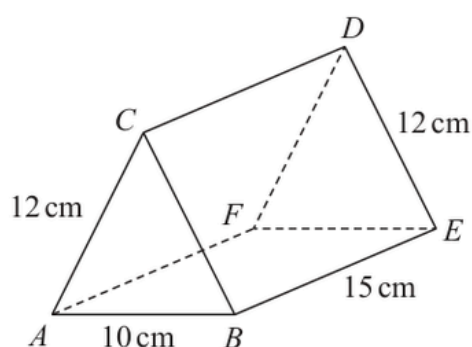


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

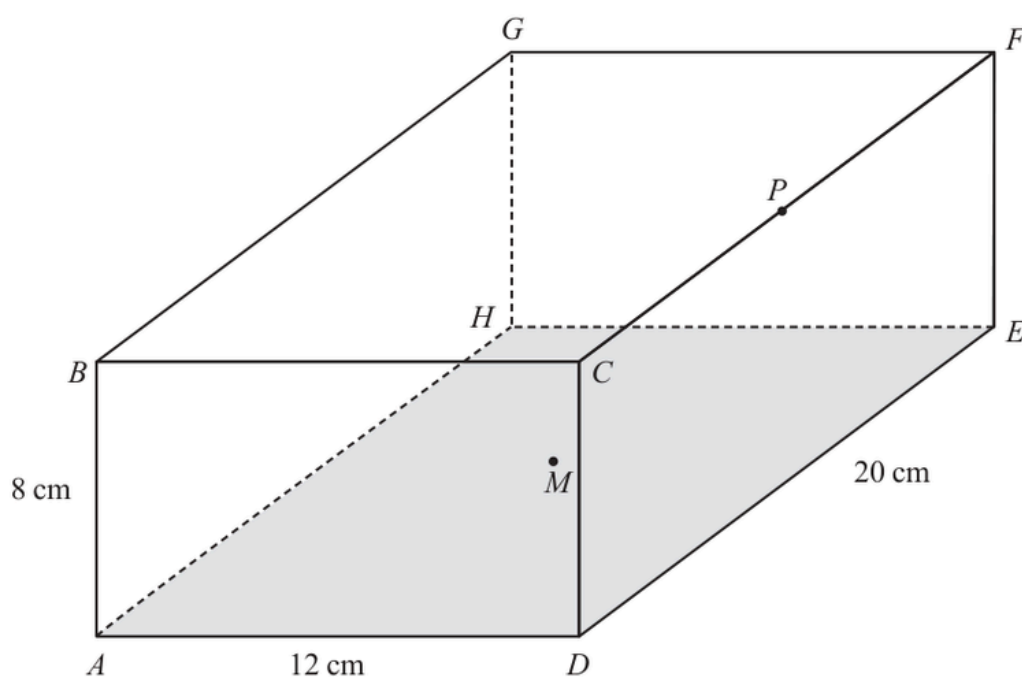
3D Pythagoras & Trigonometry

Q#: 22

Marks: 6

22 The diagram shows a cuboid $ABCDEFGH$ with horizontal base $ADEH$

Diagram NOT
accurately drawn



$$AB = 8 \text{ cm} \quad AD = 12 \text{ cm} \quad DE = 20 \text{ cm}$$

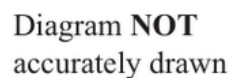
M is the midpoint of the base $ADEH$ and P is the midpoint of the edge CF

Work out the size of angle BMP

Give your answer correct to one decimal place.

(Total for Question 22 is 6 marks)

26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$



$AB = 10\text{ cm}$ $BG = 14\text{ cm}$

angle DAB = angle ABC = angle EFG = angle FGH = 90°

$$BC = \frac{1}{5}AD$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.

.....cm³

(Total for Question 26 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Histograms

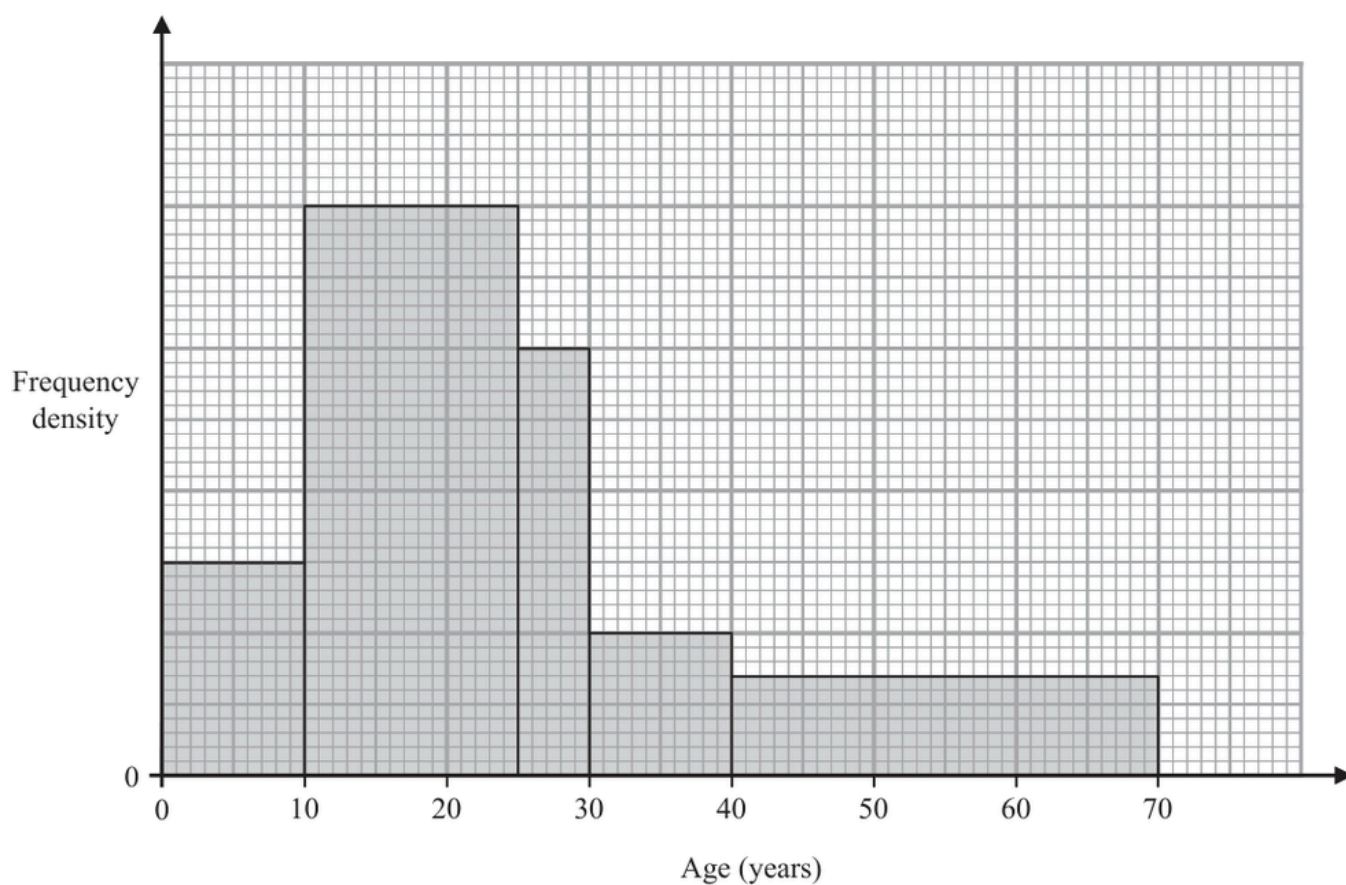
9 questions

Histograms

Q#: 21

Marks: 3

21 The histogram shows information about the ages of the members of a chess club.



There are 60 members aged between 10 and 25

There are no members older than 70

Work out an estimate for the number of members who are over 35 years of age.

(Total for Question 21 is 3 marks)

Histograms

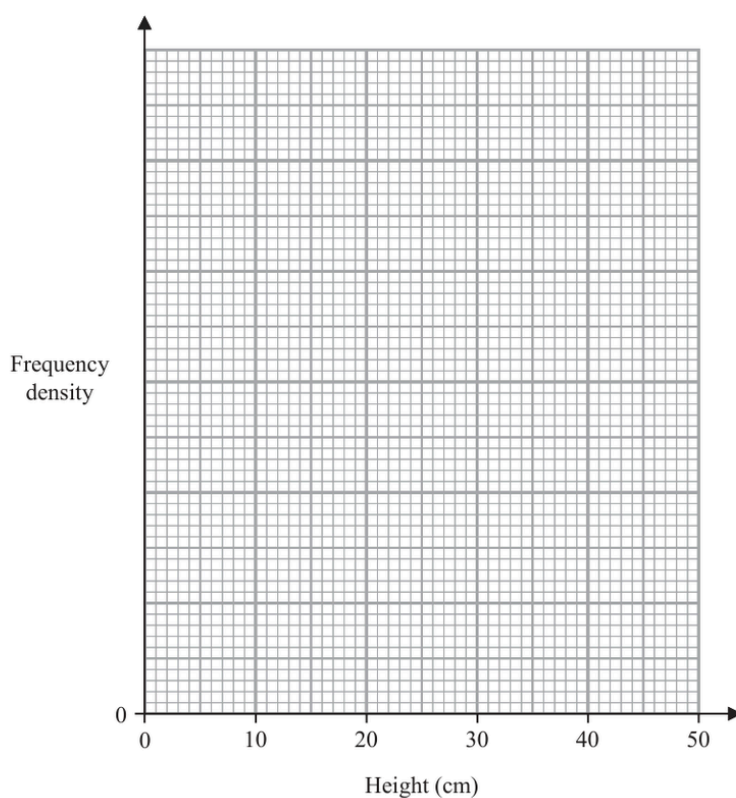
Q#: 20

Marks: 3

20 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$0 < h \leq 5$	12
$5 < h \leq 15$	34
$15 < h \leq 30$	45
$30 < h \leq 50$	28

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Probability Toolkit

8 questions

Probability Toolkit

Q#: 22

Marks: 3

22 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

(Total for Question 22 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Histograms

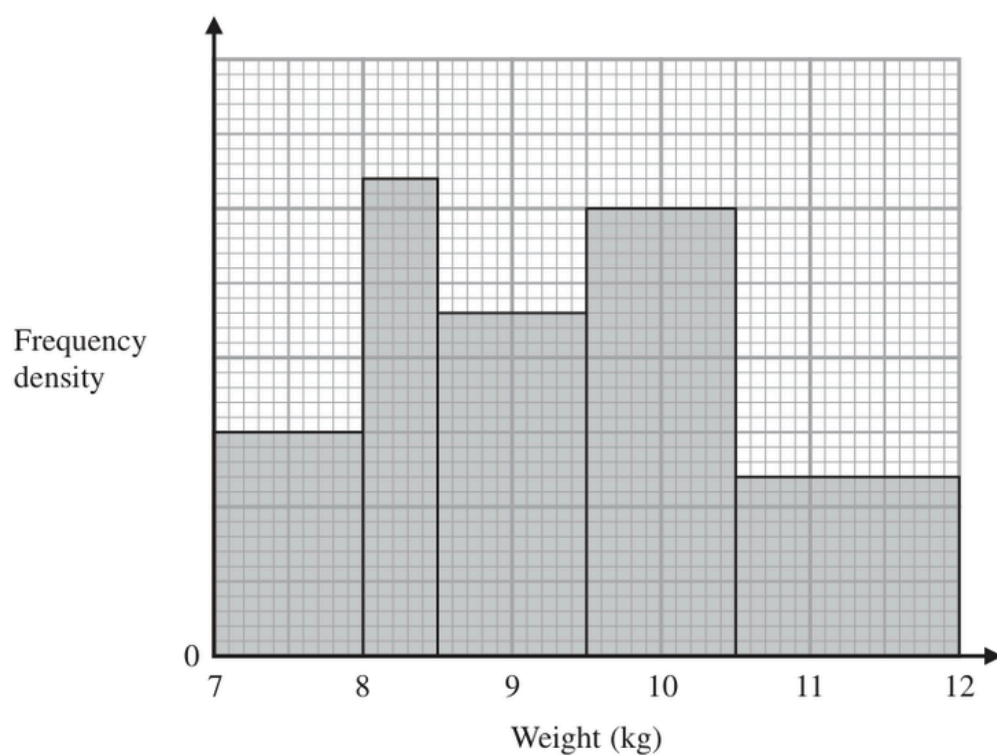
9 questions

Histograms

Q#: 21

Marks: 3

21



The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

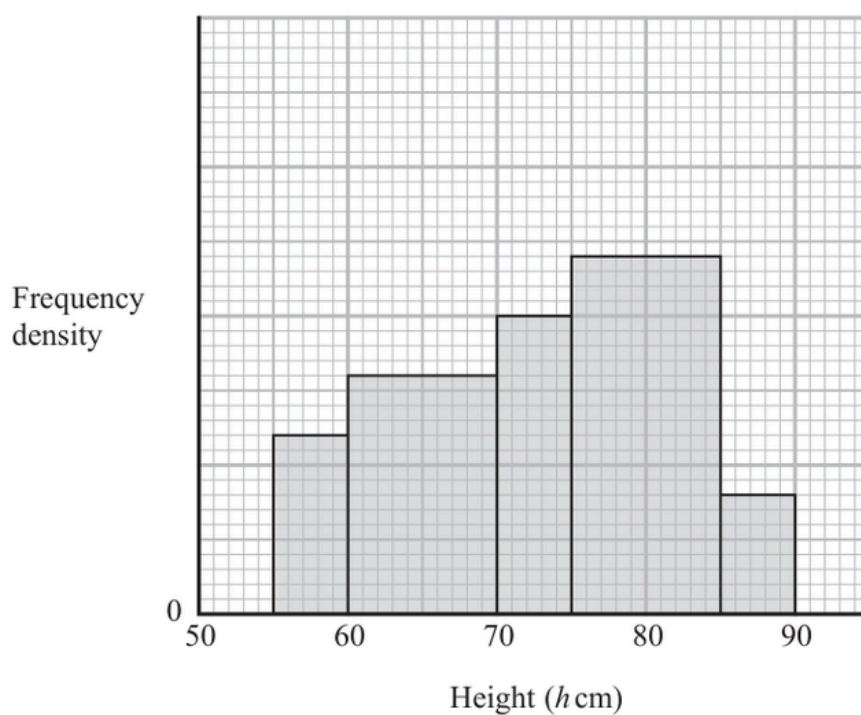
(Total for Question 21 is 3 marks)

Histograms

Q#: 20

Marks: 4

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leq 85$

One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

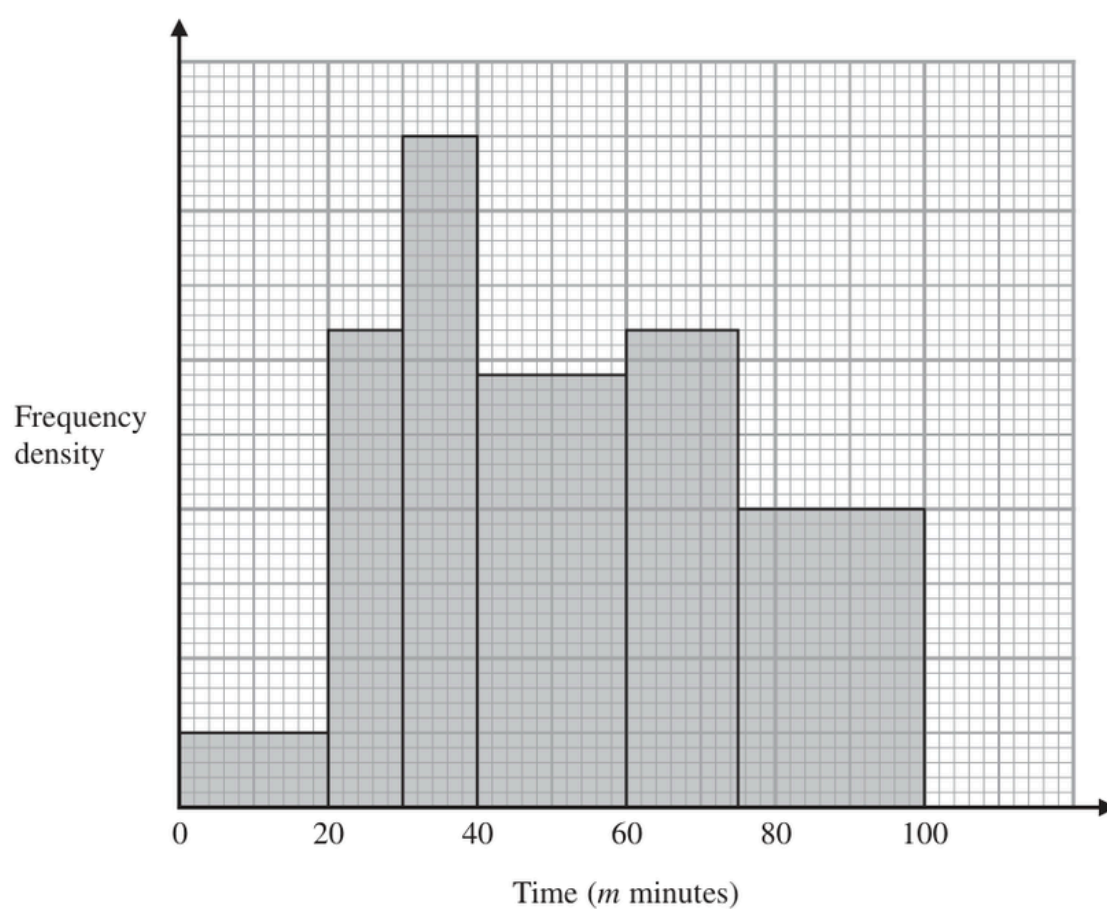
(Total for Question 20 is 4 marks)

Histograms

Q#: 21

Marks: 3

- 21** The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

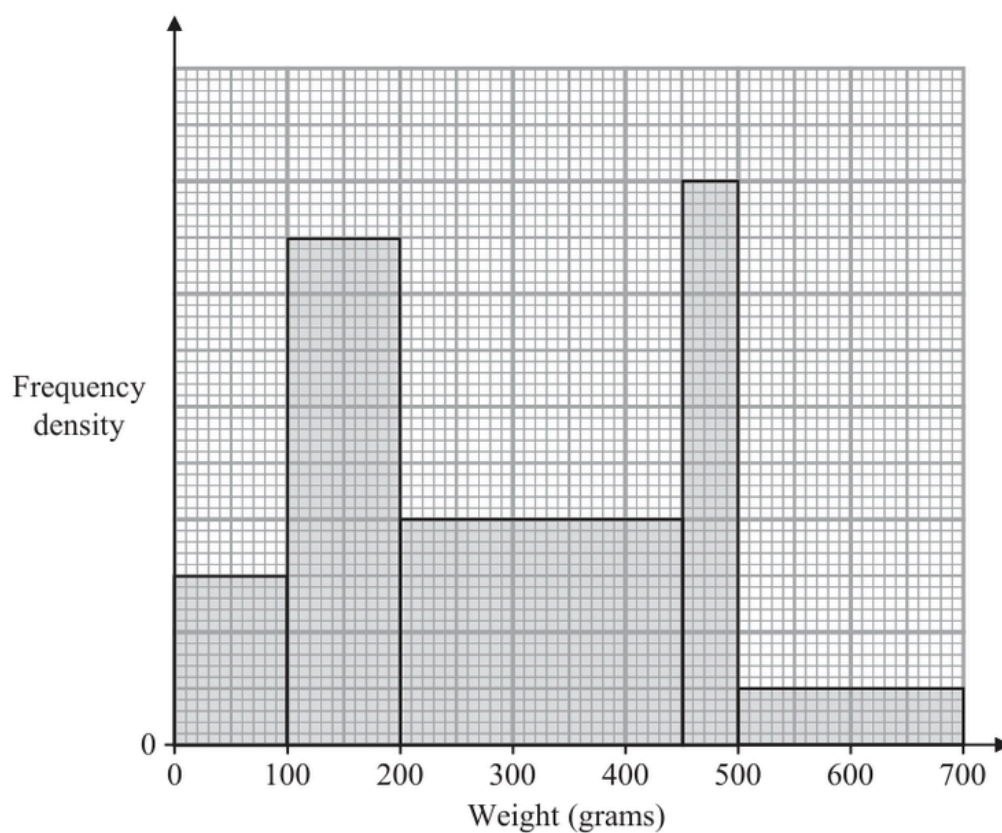
(Total for Question 21 is 3 marks)

Histograms

Q#: 20

Marks: 4

20 The histogram gives some information about the weights, in grams, of some books.



75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

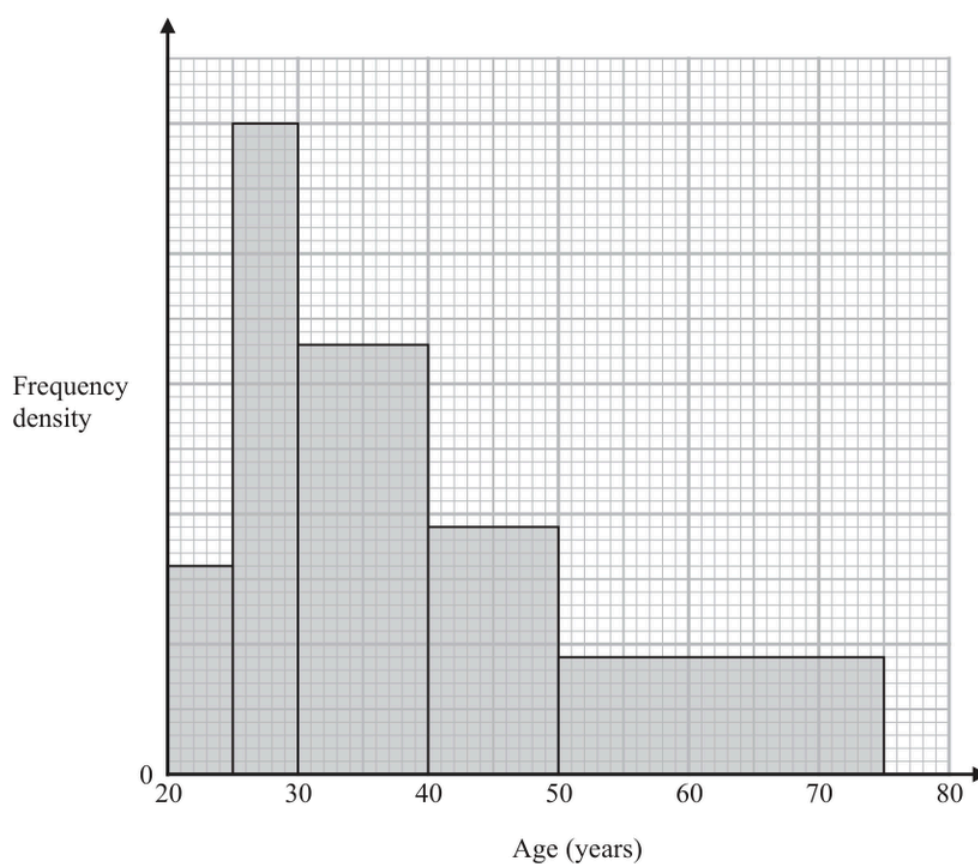
(Total for Question 20 is 4 marks)

Histograms

Q#: 22

Marks: 4

- 22 Some people attend a concert.
The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.
Give your answer correct to one decimal place.

..... %

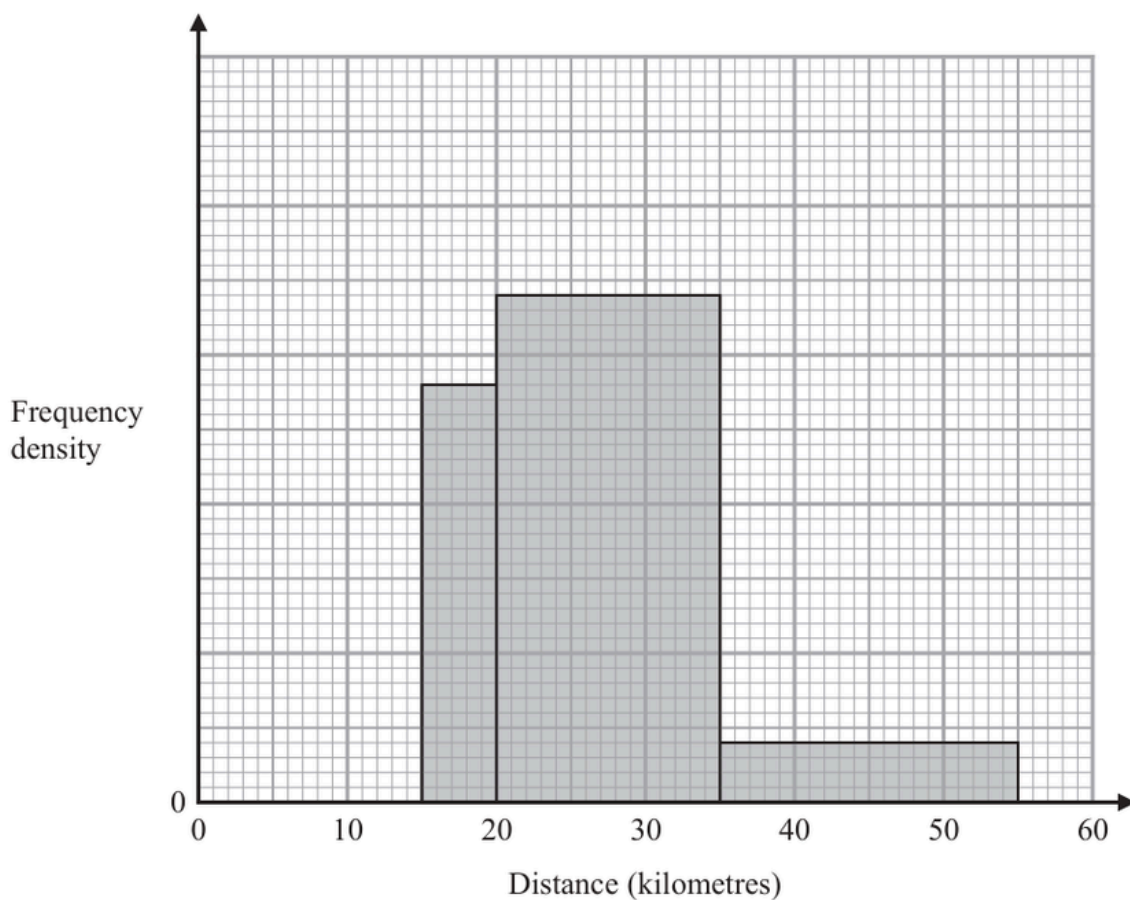
(Total for Question 22 is 4 marks)

Histograms

Q#: 22

Marks: 3

- 22** The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.
None of the adults ran more than 55 kilometres.
14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

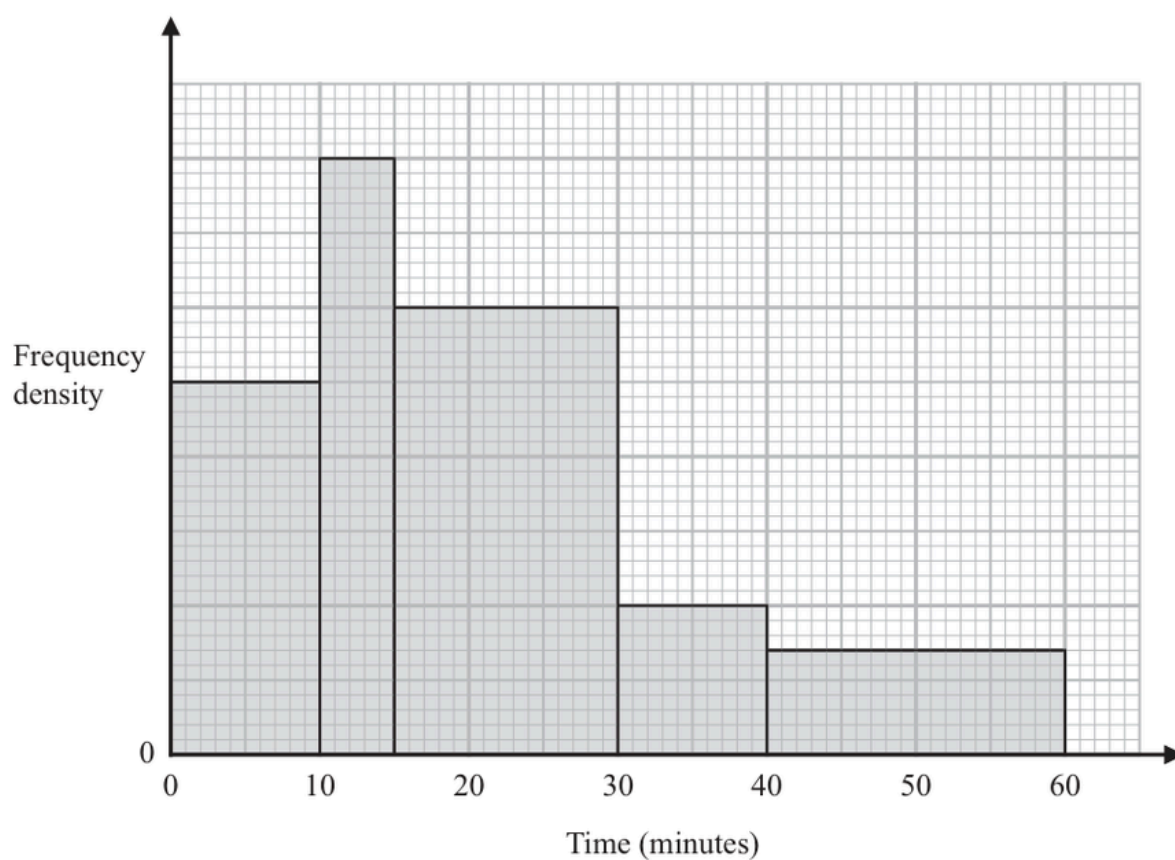
(Total for Question 22 is 3 marks)

Histograms

Q#: 21

Marks: 3

- 21** The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

(Total for Question 21 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Combined & Conditional Probability

3 questions

Combined & Conditional Probability

Q#: 23

Marks: 3

23 There are 15 buttons in a box.

- 3 of the buttons are red
- 2 of the buttons are pink
- 10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 23 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Probability Toolkit

8 questions

Probability Toolkit

Q#: 21

Marks: 6

- 21** A bag contains n beads.
6 of the beads are red and the rest are blue.

Ravi is going to take at random 2 beads from the bag.

The probability that the 2 beads will be of the same colour is $\frac{9}{17}$

Using algebra, and showing each stage of your working, calculate the value of n .

$n = \dots\dots\dots$

(Total for Question 21 is 6 marks)

Probability Toolkit

Q#: 24

Marks: 5

24 Elliot has x counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.

The probability that there are still 5 counters showing a red face is $\frac{19}{32}$

Work out the value of x
Show clear algebraic working.

$x =$

(Total for Question 24 is 5 marks)

Probability Toolkit

Q#: 21

Marks: 3

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

(Total for Question 21 is 3 marks)

Probability Toolkit

Q#: 23

Marks: 5

23 Pippa has a box containing N pens.

There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N .

Show clear algebraic working.

(Total for Question 23 is 5 marks)

Probability Toolkit

Q#: 21

Marks: 5

21 There are 25 counters in a bag such that

6 counters are blue

x counters are orange, where $x > 9$

the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.

Show clear algebraic working.

(Total for Question 21 is 5 marks)

Probability Toolkit

Q#: 23

Marks: 4

23 In a bag, there are only

3 blue beads
4 white beads
and x orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is $\frac{3}{8}$

Find the total number of beads in the bag.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

Probability Toolkit

Q#: 20

Marks: 4

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

(Total for Question 20 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Probability Diagrams - Venn & Tree Diagrams

1 questions

Probability Diagrams - Venn & Tree Diagrams

Q#: 20

Marks: 6

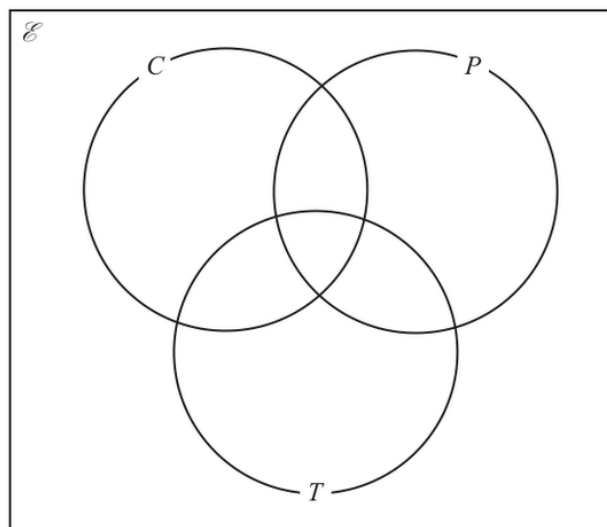
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

- 43 grow carrots
- 12 grow carrots and potatoes and tomatoes
- 18 grow carrots and potatoes
- 27 grow carrots and tomatoes
- 32 grow potatoes and tomatoes
- 29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

(Total for Question 20 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Combined & Conditional Probability

3 questions

Combined & Conditional Probability

Q#: 20

Marks: 3

20 There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

(Total for Question 20 is 3 marks)

Combined & Conditional Probability

Q#: 22

Marks: 3

22 There are 15 buttons in a box.

- 3 of the buttons are red
- 2 of the buttons are pink
- 10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)